Detecting Power Plant NO_x Emissions Changes using Physics-Guided Deep Learning and TEMPO Satellite Imagery

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Abstract

NASA's Tropospheric Emissions: Monitoring of Pollution (TEMPO) instrument provides hourly daytime observations of tropospheric nitrogen dioxide (NO_2) across North America. These observations could support independent, facility-level monitoring of power plant nitrogen oxides (NO_x) emissions, filling gaps caused by continuous emissions monitoring system (CEMS) failures and covering facilities not subject to hourly emissions-reporting requirements. Previous studies have estimated hourly point-source emissions from individual satellite scenes using empirical methods such as cross-sectional flux and the directional derivative approach. However, these estimates are sensitive to retrieval uncertainty, changing atmospheric transport, TEMPO's coarse spatial resolution, signals from nearby emitters, and temporal mismatch with hourly CEMS totals. We instead formulate facility monitoring as change detection over sequences of TEMPO observations. This relative target uses repeated scans to capture plume evolution while reducing sensitivity to persistent background levels and biases in absolute column values. We develop a physics-guided convolutional gated recurrent unit model (ConvGRU) that processes five consecutive TEMPO observations, High-Resolution Rapid Refresh (HRRR) meteorological fields, and transport residuals. The network classifies the target power plant's emissions as increasing, steady, or decreasing. On a held-out evaluation set containing more than 20,000 samples from over 95 U.S. power plants observed between August 2023 and July 2026, the model achieves a classification macro-F1 score of 0.54, which outperformed a baseline MLP (0.49). These results support the use of sequential TEMPO imagery for detecting short-term changes in power plant emissions across facilities and observing conditions.

1 Introduction

Based on U.S. Environmental Protection Agency (EPA) inventory data, power plants in the U.S. electricity-generation sector emitted approximately 0.74 million short tons of NO_x in 2023, about 9.5% of national emissions [13]. NO_x emissions contribute to ground-level ozone and secondary particulate matter, pollutants that can inflame and damage the airways, reduce lung function, and aggravate asthma [11]. EPA regulations require most large U.S. fossil-fuel-fired generating units that produce electricity for sale to submit unit-level hourly emissions data on a quarterly basis [14]. However, monitoring equipment malfunctions and maintenance can make CEMS measurements unavailable or invalid, requiring plants to report calculated substitute values instead of direct measurements [12]. Behind-the-meter generators that do not sell electricity can fall outside federal hourly reporting requirements, as illustrated by the natural-gas turbines powering SpaceXAI's Colossus 2 data center [5]. Outside the United States, comparable public hourly, unit-level emissions data are generally unavailable, even where facilities continuously monitor emissions [6]. Independent estimates could therefore fill reporting gaps and reveal short-term changes before the EPA's quarterly release of hourly emissions data.

TEMPO is an ultraviolet-visible spectrometer hosted aboard a geostationary satellite that observes tropospheric vertical column densities (TVCDs) for various trace gases across North America during daylight hours [16]. Earlier polar-orbiting instruments, including the Ozone Monitoring Instrument (OMI) and TROPOspheric Monitoring Instrument (TROPOMI), typically observe a given location once per day; TEMPO's geostationary view provides approximately hourly scans with finer spatial resolution [7, 15, 16]. Hourly sampling captures intraday changes in power plant plumes and atmospheric transport that daily observations can miss. However, power plant signals in TEMPO NO_2 TVCDs can be weakened or obscured by factors including evolving meteorology and atmospheric chemistry, cloud cover, viewing geometry and retrieval uncertainty, dilution within TEMPO's 2.0-by-4.75 km spatial resolution per pixel, and interference from nearby emission sources [5, 6, 10]. Cross-comparisons with CEMS data, which is the most reliable ground truth available, introduce an additional temporal mismatch because TEMPO observes a facility at a specific time during each hourly scan, whereas CEMS data aggregate emissions over an entire clock hour [10, 14].

Satellite-based point-source studies typically combine observed column enhancements with modeled atmospheric transport. Cross-sectional flux (CSF) integrates background-subtracted columns across plume transects, while the directional derivative approach (DDA) relates spatial column gradients to wind fields [8, 9]. Santaren et al. benchmarked five such methods on synthetic daily satellite scenes with known source rates and found that cloud gaps and wind errors limited their accuracy and coverage [8]. For TEMPO, Sun et al. added hourly changes in the observed NO_2 column to DDA and compared estimates for 14 U.S. power plants with CEMS, but retrieval and chemical-conversion uncertainties produced substantial low bias [9]. Beaudry et al. fit a rise–decay model to TEMPO plumes from 64 plants and also found weaker performance for lower-emitting sources [2].

Machine learning offers an alternative by learning emission-related plume patterns from data. Huang and Wang predicted whether TROPOMI could detect power plant NO_2 plumes from plant, meteorological, surface, and sensor variables rather than estimating emissions from the imagery [6]. Alnaim et al. used recurrent models with TROPOMI, meteorological, and EPA data to estimate power plant emissions, but did not test transfer to previously unseen plants [1]. Regression has also been demonstrated for

other gases: U-Plume combines learned methane plume masks with imagery and wind to infer source rates, while Dumont et al. trained a CNN on simulated CO_2 and wind fields to estimate power plant emissions [3, 4]. To our knowledge, no published study has used sequences of hourly TEMPO NO_2 imagery to infer facility-level emissions or behavior.

Estimating absolute hourly emissions from individual TEMPO scenes requires separating weak facility signals from background pollution and resolving uncertainties in retrieval, atmospheric chemistry, and transport. We instead use TEMPO's hourly sampling to detect relative changes over short observation sequences, reducing dependence on absolute column values and persistent background levels.

Our contributions are as follows:

- We formulate satellite-based power plant monitoring as a three-class change-detection task that identifies increasing, steady, and decreasing emissions behavior.
- We introduce a physics-guided deep learning framework that combines sequences of TEMPO NO_2 observations with meteorology and transport information to predict emissions behavior.
- We evaluate the model on held-out power plants across geographic regions, seasons, and operating profiles and compare its performance with the temporal MLP.

2 Data Sources

2.1 TEMPO NO_2 Data

NASA distributes TEMPO NO_2 retrievals in two formats: Level 2 data on the instrument's native measurement footprints and Level 3 data regridded from each east-to-west scan [10]. We downloaded version 4 Level 2 NO_2 retrievals from August 2023 through August 2026. Along with NO_2 TVCDs, the product provides metadata including native footprint geometry, mirror-step observation times, main data quality flags, and effective cloud fractions, which we later use for regridding, quality filtering, and temporal alignment with CEMS records.

2.2 CEMS Records

We downloaded hourly emissions records for individual power plant units and annual facility and unit attributes from the U.S. Environmental Protection Agency's Clean Air Markets Program Data (CAMPD) APIs for August 2023 through September 2026. From the hourly records, we use facility and unit identifiers, reporting date and hour, NO_x mass and its measurement flag, operating time, and primary fuel. From the annual attributes, we use facility coordinates and unit-level primary fuel.

2.3 HRRR Weather Data

The High-Resolution Rapid Refresh (HRRR) is an hourly weather model with a 3 km grid covering the contiguous United States. From NOAA's public HRRR archive, we downloaded hourly 2 m air temperature and eastward and northward wind components at 80 m above ground for August 2023 through August 2026.

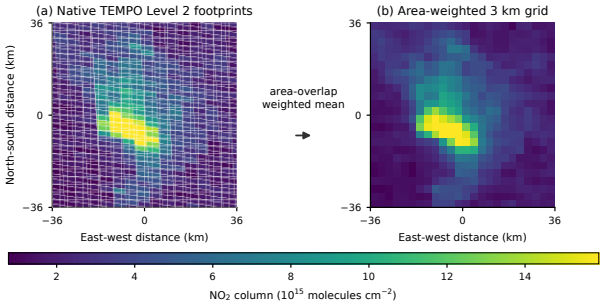


Figure 1: TEMPO Level 2 regridding to the AOI grid.

3 Methods

3.1 AOI Construction

For each power plant, we define a 72 km × 72 km area of interest (AOI) centered on the facility and divide it into a 24 × 24 grid of 3 km × 3 km cells. For each AOI-hour, we sum the unit-level NO_x mass reported by all facilities in the AOI and discard the hour if any contributing measurement is invalid or unavailable. We score each AOI using its median total NO_x mass across hours whose mean unit operating time meets or exceeds the AOI-specific median. We retain the highest-scoring 50% to focus the dataset on emissions signals more likely to be detected by TEMPO.

3.2 TEMPO Regridding and Quality Filtering

TEMPO Level 2 observations lie on irregular native footprints, whereas our model uses a fixed spatial grid. We project the footprint boundaries and AOI grid to the contiguous-U.S. Albers equal-area coordinate system. We then assign each AOI grid cell an NO_2 TVCD by averaging the TEMPO footprints that intersect it, weighted by their area of overlap. Figure 1 illustrates this mapping from native footprints to the fixed AOI grid.

A footprint contributes to the NO_2 average only when its main data quality flag equals zero and its effective cloud fraction does not exceed 0.20. Grid cells without an accepted footprint overlap remain masked, and we retain a binary mask of the original valid cells. We fill the masked NO_2 cells by bilinear interpolation before later processing.

3.3 Sample Construction

Each sample contains five consecutive observations of one AOI, indexed from t_0 to t_4. We retain a sequence only when each pair of adjacent TEMPO scans is separated by 40–70 minutes and every NO_2 raster has valid pixel coverage of at least 85%. At each timestep, we pair the TEMPO raster with the HRRR temperature and wind rasters from the closest hour interpolated to a common 24 × 24 raster size and spatial coverage as the NO_2 rasters.

Because CEMS records correspond to clock hours rather than TEMPO scan times, we linearly interpolate the AOI-hour NO_x emissions between the surrounding hourly CEMS records at each scan time. Let E_i denote the interpolated emissions at time t_i. For

scans t_a through t_b, we calculate the time-weighted mean

$$\overline{E}_{a:b} = \frac{\sum_{i=a}^{b-1}(t_{i+1} - t_i)(E_i + E_{i+1})/2}{t_b - t_a}. \tag{1}$$

We define the sequence change as $\Delta E = \overline{E}_{2:4} - \overline{E}_{0:2}$ and use the threshold $\theta = 250\ \mathrm{lb\,h^{-1}}$ to assign its category:

$$c = \begin{cases} \text{increase}, & \Delta E \geq \theta, \\ \text{steady}, & -\theta < \Delta E < \theta, \\ \text{decrease}, & \Delta E \leq -\theta. \end{cases} \tag{2}$$

Figure 2 shows one complete five-timestep model input and its derived behavior label.

3.4 Transport Model

We also supply the model with rasters that use a transport model to separate new NO_2 emissions from existing plume at a given timestep. For each interval $\Delta t_i = t_{i+1} - t_i$, let $N_i(\boldsymbol{x})$ denote the earlier NO_2 raster, $\boldsymbol{u}_i(\boldsymbol{x})$ its wind field, and B_i the median NO_2 value along the raster boundary. We approximate transport as

$$\begin{aligned} \overline{\boldsymbol{u}}_i &= \gamma\left[(1-\lambda)\boldsymbol{u}_i + \lambda \boldsymbol{u}_{i+1}\right], \\ A_i(\boldsymbol{x}) &= N_i(\boldsymbol{x} - \overline{\boldsymbol{u}}_i(\boldsymbol{x})\Delta t_i), \\ \sigma_i &= \sigma_0\sqrt{\frac{\Delta t_i}{1\ \mathrm{h}}}, \\ D_i &= B_i + G_{\sigma_i} * (A_i - B_i), \\ \widehat{N}_{i+1} &= B_i + \exp\left(-\frac{\Delta t_i}{\tau}\right)(D_i - B_i), \\ R_i &= N_{i+1} - \widehat{N}_{i+1}. \end{aligned} \tag{3}$$

The semi-Lagrangian step backtraces each destination cell and uses bilinear sampling to obtain the advected field A_i. Convolution with the Gaussian kernel G_{σ_i} disperses the enhancement above the boundary background, and the exponential term applies effective NO_2 decay with lifetime τ. The parameters γ, λ, σ_0, and τ control wind-speed scaling, endpoint-wind blending, dispersion, and decay; we select them using validation data as part of the model optimization procedure described later.

We compute one residual R_i for each adjacent pair, producing four residual rasters per sample. We transport the earlier validity mask through the same advection and dispersion steps and combine it with the later mask to identify unreliable residual pixels. The transport-residual panel in Figure 2 shows the four adjacent-scan residuals for the same example sequence.

3.5 Dataset Stratification

Our split design has three goals: prevent leakage from overlapping AOIs, approximate a 70%/15%/15% division, and distribute the three classes across the splits while limiting class imbalance. We group overlapping AOIs into connected clusters and treat each cluster as an indivisible assignment unit. Because each cluster contains an arbitrary number of AOIs, samples, and class instances, cluster-level assignment cannot generally meet every target exactly. Steady samples account for $\sim$ 90% of the raw sample pool, compared with $\sim$ 7% increase and $\sim$ 3% decrease samples. Although we later use a class-weighted training loss, we control the composition of each split so that steady behavior does not dominate training or evaluation.

We use mixed-integer programming to assign each cluster to exactly one split while minimizing normalized deviations from the target AOI and sample counts. The first optimization targets the 70%/15%/15% division and fixes the resulting training clusters. A second optimization divides the remaining clusters equally between validation and test while balancing their AOI, sample, and per-class counts. We then downsample each training class to the size of the smallest class. In validation and test, we retain every increase and decrease sample and uniformly sample steady examples until their count matches that of the more common change class.

3.6 ConvGRU Model

We use a convolutional recurrent neural network with a convolutional gated recurrent unit (ConvGRU) to preserve spatial structure while combining the five observations in each sample. At each timestep, an observation encoder processes NO_2, the NO_2 quality mask, temperature, and the two wind components. A separate encoder processes each of the four transport residuals together with its quality mask. Each encoder shares weights across time and reduces the 24×24 inputs to spatial feature maps for temporal aggregation. Figure 3 summarizes the architecture.

The encoded first observation initializes the recurrent hidden state. At each later timestep, the model concatenates the current observation features with the transport residual from the preceding interval, fuses them with a 1×1 convolution, and updates the hidden state with the ConvGRU. Its convolutional gates retain the spatial arrangement of plume features while learning how much information to preserve or replace across the sequence. This design lets the network use both the observed atmospheric state and the portion of each change that the transport model does not explain.

The final recurrent feature map feeds the classification head. We combine global average and maximum pooling to form a visual embedding. A small feedforward network separately encodes the four temporal features, and its output is concatenated with the visual embedding before the classification head produces logits for the increase, steady, and decrease classes. This late-fusion design keeps the tabular context separate while the ConvGRU learns the raster sequence.

3.7 Temporal MLP

The temporal MLP measures how much emissions-change behavior can be predicted from recurring daily and annual patterns without satellite or weather inputs. We represent each sample using four cyclic features: the sine and cosine of local solar hour (midpoint of timesteps) and the sine and cosine of day of year. The temporal MLP maps these features through hidden layers of 16 and 8 units, each followed by layer normalization and a SiLU activation. A linear output layer produces logits for the three behavior classes. Figure 4 shows this architecture.

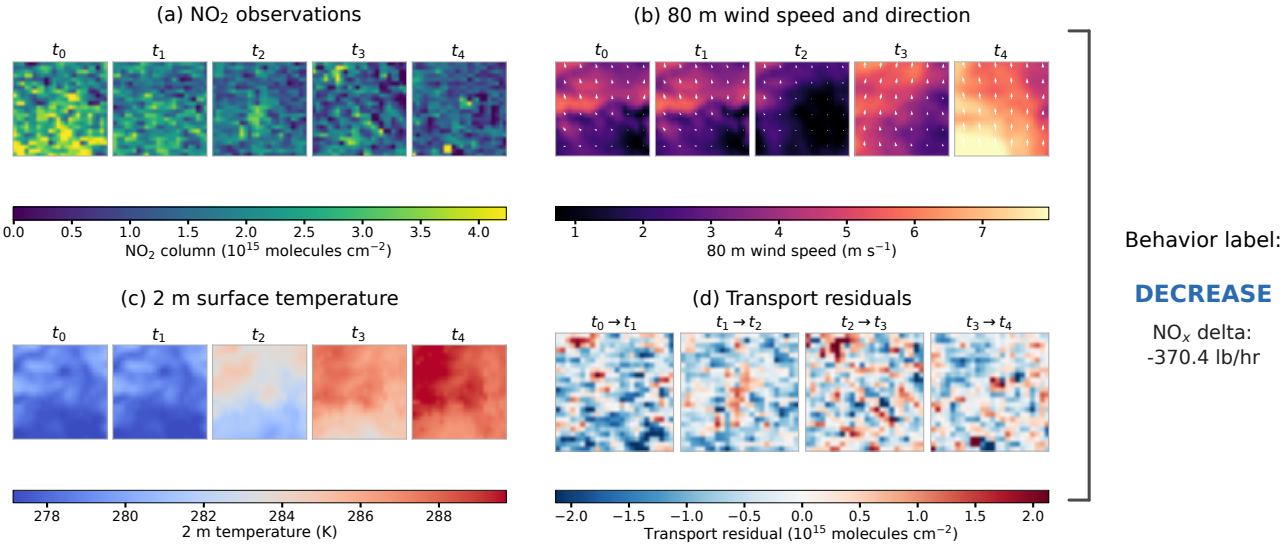


Figure 2: Complete model inputs and behavior label for one five-timestep training sample.

Placeholder: ConvGRU architecture

Raster encoders → ConvGRU → visual embedding
Temporal features → feedforward encoder → late fusion → classification head

Figure 3: ConvGRU architecture.

Placeholder: baseline MLP

Four cyclic time features → baseline MLP → three class logits

Figure 4: Temporal MLP architecture.

Table 1: Dataset composition by split and class.

Split	AOIs	Decrease	Steady	Increase	Total
Training	469	17,130	23,289	29,888	70,307
Validation	101	3,198	8,589	10,411	22,198
Test	97	3,184	7,764	10,093	21,041

3.8 Training Procedure

Table 1 summarizes the final dataset used to train and evaluate both models. We standardize the NO_2, wind, temperature, and transport-residual rasters and the tabular inputs using the corresponding training-set mean and standard deviation. We apply these training-derived statistics unchanged to the validation and test splits.

Both classifiers use inverse-frequency class weights computed from the final training split and minimize weighted cross-entropy.

4 Results

4.1 Classification Performance

We evaluate classification using macro-F1 and balanced accuracy to account for differences in class prevalence. On the test set, the ConvGRU achieves a macro-F1 of 0.538 and balanced accuracy of 0.553, compared with 0.494 and 0.513 for the temporal MLP. As shown in Figure 5, the ConvGRU improves decrease and steady classification, while its increase scores are lower than the temporal MLP's on the test split. Results across the training, validation, and test splits show the same aggregate advantage for the ConvGRU.

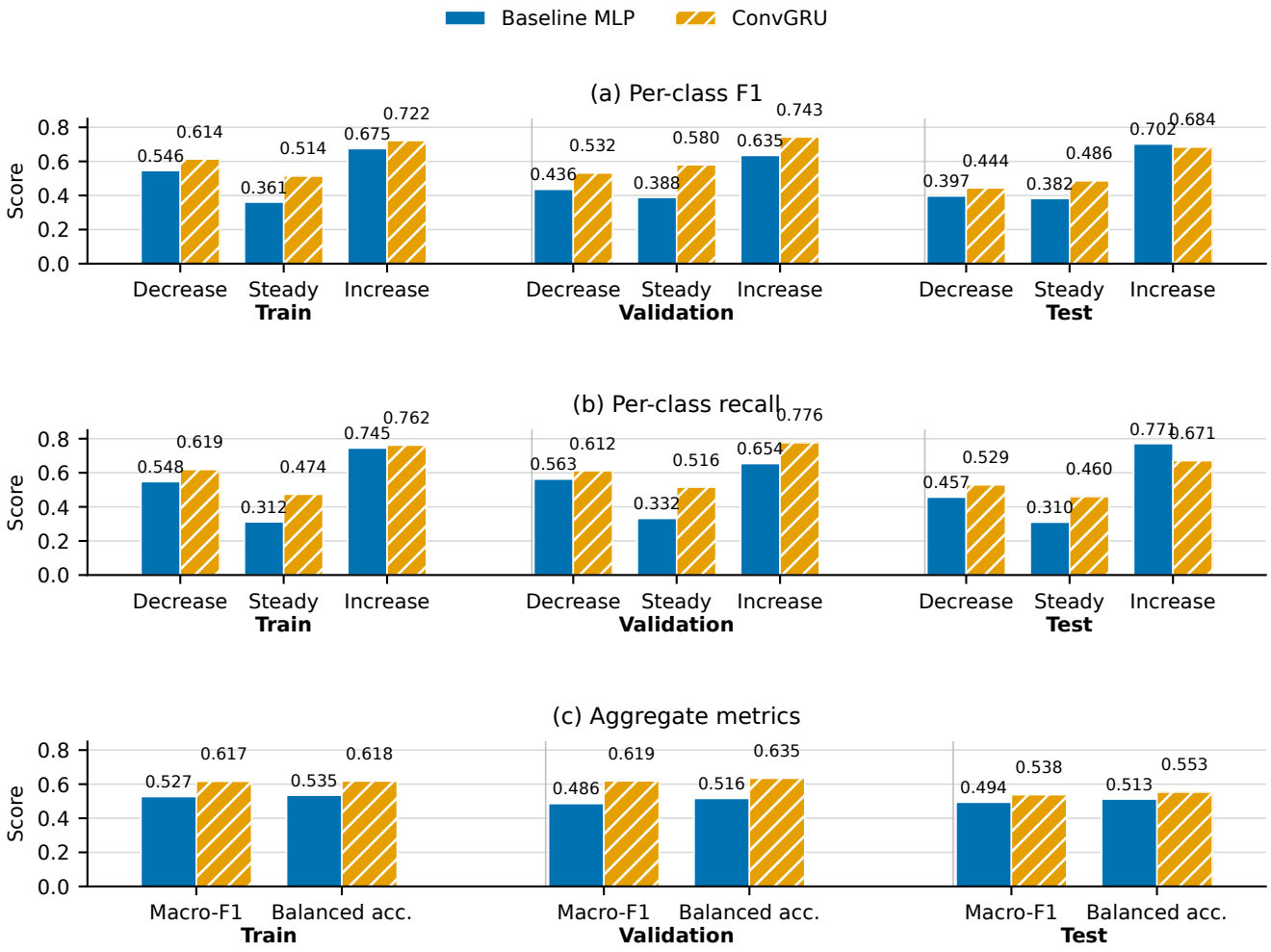


Figure 5: Classification performance by split.

Table 2: Training configuration.

Parameter	Baseline MLP	ConvGRU
Trainable parameters	291	724,939
Optimizer	AdamW	
Batch size	128	128
Initial learning rate	10^{-3}	6.04×10^{-5}
Weight decay	10^{-4}	10^{-4}
Dropout	–	0.20
Gradient-norm limit	5.0	
LR scheduler	Reduce on plateau	
LR reduction factor	0.5	
Scheduler patience	5 epochs	
Selection metric	Validation macro-F1	
Early-stopping patience	10 epochs	
Hardware	1 NVIDIA RTX A5000 GPU	

4.2 Performance Across Evaluation Strata

4.3 Ablation Study

References

[1] Ahmed Alnaim, Ziheng Sun, and Daniel Tong. 2022. Evaluating Machine Learning and Remote Sensing in Monitoring NO_2 Emission of Power Plants. *Remote Sensing* 14, 3 (2022), 729. https://doi.org/10.3390/rs14030729

[2] Samuel James Beaudry, Qindan Zhu, and Ronald C. Cohen. 2026. Quantification of Power Plant NO_x Emissions: A Reference for Diurnal and Seasonal Accuracy of TEMPO. https://doi.org/10.31223/X5B808 EarthArXiv preprint.

[3] Jack H. Bruno, Dylan Jervis, Daniel J. Varon, and Daniel J. Jacob. 2024. U-Plume: Automated Algorithm for Plume Detection and Source Quantification by Satellite Point-Source Imagers. *Atmospheric Measurement Techniques* 17 (2024), 2625–2636. https://doi.org/10.5194/amt-17-2625-2024

[4] J. Dumont Le Brazidec, P. Vanderbecken, A. Farchi, G. Broquet, G. Kuhlmann, and M. Bocquet. 2024. Deep Learning Applied to CO_2 Power Plant Emissions Quantification Using Simulated Satellite Images. *Geoscientific Model Development* 17 (2024), 1995–2014. https://doi.org/10.5194/gmd-17-1995-2024

[5] Kevin D. Gauld, Daniel J. Varon, Nicholas Balasus, and Daniel H. Cusworth. 2026. Quantifying AI Data Center Nitrogen Oxide (NO_x) Emissions from Space. https://doi.org/10.48550/arXiv.2608.22153 arXiv:2608.22153 [physics.ao-ph]

[6] Ruizhe Huang and Sherrie Wang. 2026. Global Variability in the Detectability of Power Plant NO_2 Plumes from Space. *Atmospheric Measurement Techniques* 19 (2026), 6099–6124. https://doi.org/10.5194/amt-19-6099-2026

[7] Pieternel F. Levelt, Gijsbertus H. J. van den Oord, Marcel R. Dobber, Anssi Mälkki, Huib Visser, Johan de Vries, Piet Stammes, Jens O. V. Lundell, and Heikki Saari. 2006. The Ozone Monitoring Instrument. *IEEE Transactions on Geoscience and Remote Sensing* 44, 5 (2006), 1093–1101. https://doi.org/10.1109/TGRS.2006.872333

[8] D. Santaren, J. Hakkarainen, G. Kuhlmann, E. Koene, F. Chevallier, I. Ialongo, H. Lindqvist, J. Nurmela, J. Tamminen, L. Amorós, D. Brunner, and G. Broquet. 2025. Benchmarking Data-Driven Inversion Methods for the Estimation of Local CO_2 Emissions from Synthetic Satellite Images of XCO_2 and NO_2. *Atmospheric Measurement Techniques* 18 (2025), 211–239. https://doi.org/10.5194/amt-18-211-2025

[9] Kang Sun, Jobaer Ahmed Saju, Caroline R. Nowlan, Gonzalo González Abad, and Xiong Liu. 2025. Hourly Nitrogen Oxides Emissions Estimated From TEMPO and Comparison With Facility-Level Monitoring Data. *Journal of Geophysical Research: Atmospheres* 130, 23 (2025), e2025JD044565. https://doi.org/10.1029/2025JD044565

[10] TEMPO Science Team. 2026. *TEMPO Trace Gas and Cloud Level 2 and 3 Data Products: User Guide.* NASA Langley Atmospheric Science Data Center. https://doi.org/10.5067/doc/tempo/L2andL3UserGuide/V2.1 Version 2.1.

[11] U.S. Environmental Protection Agency. 2016. *Integrated Science Assessment for Oxides of Nitrogen—Health Criteria.* Technical Report EPA/600/R-15/068. U.S. Environmental Protection Agency. https://www.epa.gov/isa/integrated-science-assessment-isa-oxides-nitrogen

[12] U.S. Environmental Protection Agency. 2021. *Substitute Data in EPA CAMD's Power Sector Emissions Data.* Technical Report. U.S. Environmental Protection Agency. https://www.epa.gov/system/files/documents/2021-06/monitoring-insights-substitute-data.pdf

[13] U.S. Environmental Protection Agency. 2026. Air Pollutant Emissions Trends Data. https://www.epa.gov/air-emissions-inventories/air-pollutant-emissions-trends-data Accessed October 4, 2026.

[14] U.S. Environmental Protection Agency. 2026. Emissions Monitoring and Reporting. https://www.epa.gov/power-sector/emissions-monitoring-and-reporting Accessed October 4, 2026.

[15] J. P. Veefkind, I. Aben, K. McMullan, H. Förster, J. de Vries, G. Otter, J. Claas, H. J. Eskes, J. F. de Haan, Q. Kleipool, M. van Weele, O. Hasekamp, R. Hoogeveen, J. Landgraf, R. Snel, P. Tol, P. Ingmann, R. Voors, B. Kruizinga, R. Vink, H. Visser, and P. F. Levelt. 2012. TROPOMI on the ESA Sentinel-5 Precursor: A GMES Mission for Global Observations of the Atmospheric Composition for Climate, Air Quality and Ozone Layer Applications. *Remote Sensing of Environment* 120 (2012), 70–83. https://doi.org/10.1016/j.rse.2011.09.027

[16] Peter Zoogman, Xiong Liu, R. M. Suleiman, W. F. Pennington, David E. Flittner, Jassim A. Al-Saadi, Brian B. Hilton, Dennis K. Nicks, Michael J. Newchurch, J. L. Carr, Scott J. Janz, M. R. Andraschko, Antti Arola, Barry D. Baker, Brian P. Canova, Christopher Chan Miller, Ronald C. Cohen, J. E. Davis, M. E. Dussault, David P. Edwards, Jack Fishman, A. Ghulam, Gonzalo González Abad, Michel Grutter, Jay R. Herman, J. Houck, Daniel J. Jacob, Joanna Joiner, Brian J. Kerridge, Jhoon Kim, Nickolay A. Krotkov, Lok N. Lamsal, Can Li, Anders Lindfors, Randall V. Martin, C. Thomas McElroy, Chris McLinden, Vijay Natraj, Daniel O. Neil, Caroline R. Nowlan, E. J. O'Sullivan, Paul I. Palmer, R. Bradley Pierce, Mark R. Pippin, Alfonso Saiz-Lopez, Robert J. D. Spurr, James J. Szykman, Omar Torres, J. Pepijn Veefkind, Ben Veihelmann, Hui Wang, Jun Wang, and Kelly Chance. 2017. Tropospheric Emissions: Monitoring of Pollution (TEMPO). *Journal of Quantitative Spectroscopy and Radiative Transfer* 186 (2017), 17–39. https://doi.org/10.1016/j.jqsrt.2016.05.008